

Integrating Large Language Models into the Chemistry and Materials Science Laboratory Curricula



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From robotic teachers to interactive tomes, personalized, artificial intelligence (AI)-based education has been a recurring theme in science fiction since the mid-20th century. With the recent rise of large language models (LLMs) such as ChatGPT and Gemini, these once-fantastical visions are rapidly shifting from fiction to fact. In the realm of chemistry and materials science education—and especially in courses with a laboratory component—the potential for revolutionizing education is immense. Laboratory experiences are meant to foster hands-on skills, reinforce core concepts, and develop students' identity as scientists.^{1–4} Yet research consistently finds that traditional lab courses underperform in achieving deeper conceptual understanding and gains in critical thinking.^{2,3,5–7} In an era of LLMs, one can envision improving these student outcomes using AI-based feedback or personalized guidance to reshape laboratory courses. Alongside cautious optimism, however, higher education faces unsettling questions surrounding ethical usage and the penchant of LLMs to deliver hallucinations and misinformation. In lab-based courses, these risks become particularly fraught.

Presently, the intersection of education and LLMs is a “wild west”, where individual educators are experimenting with LLMs in classrooms and forging new approaches as they go. Best practices remain *ad hoc*, and new strategies—whether brilliant or flawed—are rarely shared widely. The result is a patchwork of innovation that calls for a more coherent framework and a better approach to sharing our successes and failures. In an early effort to answer this call, we convened a two-day workshop, “Integrating Large Language Models into the Chemistry Laboratory Curriculum”, supported by the National Science Foundation (NSF). Our goal was to bring together a diverse group of graduate students, post-doctoral scholars, and faculty (41 participants, in total) with expertise spanning materials chemistry, chemical education, and computer science—each offering perspectives on laboratory teaching, undergraduate research experiences, and developing custom LLM-based tools. We spent much of the first half of the workshop sharing our prior experiences in areas ranging from evidence-based lab pedagogy, principles behind LLMs, and early attempts at incorporating LLMs in our courses. The second half was organized as a “hackathon”, where we worked in groups to devise novel ways of integrating LLMs into chemistry education to ultimately create virtual laboratory teaching assistants or personalized tutoring systems. Because each team included both instructors armed with existing instructional resources and computer scientists with fluency in LLMs, we made surprisingly rapid progress in the short time

frame. Within the workshop, the discussion was predominantly focused on chemistry laboratory classes (e.g., general chemistry, organic, physical), as they are some of the most common science laboratory courses taken across disciplines. Materials science students in particular benefit from chemistry courses as there is significant disciplinary overlap. Together, we grappled with several recurring questions that have emerged with the expanding influence of LLMs in chemistry education:

1. How might LLMs impact student learning in laboratory settings, especially regarding critical thinking, core concepts, and scientific identity?
2. What are the ramifications of LLMs for course delivery strategies, including course design, scaffolding, and assessment?
3. Can LLMs facilitate the incorporation of alternative delivery strategies into the undergraduate curriculum, such as course-based undergraduate research experiences (CURES), to potentially broaden access to authentic research?

1. HOW MIGHT LLMs IMPACT STUDENT LEARNING?

On the first morning of the workshop, we began by reviewing literature on the purpose and efficacy of current instructional laboratories, a foundational step before examining how LLMs might shape future learning. Laboratory courses are traditionally meant to reinforce lecture content while cultivating practical, transferable skills—for instance, data analysis, written communication, and safe laboratory practices.^{1–4} A common critique is that laboratory experiences tend to be too prescriptive, based on “cookbook” protocols that have been developed and refined to be safe and time-effective.^{2,3,5–7} Although this structure still provides benefits (e.g., tactile experience using laboratory equipment and instrumentation), some students come away from their undergraduate education with an unrealistic—or worse, a negative—opinion of laboratory science and research.^{3,6,8} An ideal laboratory format would reflect the nature of the scientific process, encouraging scientific inquiry, conceptual understanding, and critical thinking. However, this requires reforms to the entire

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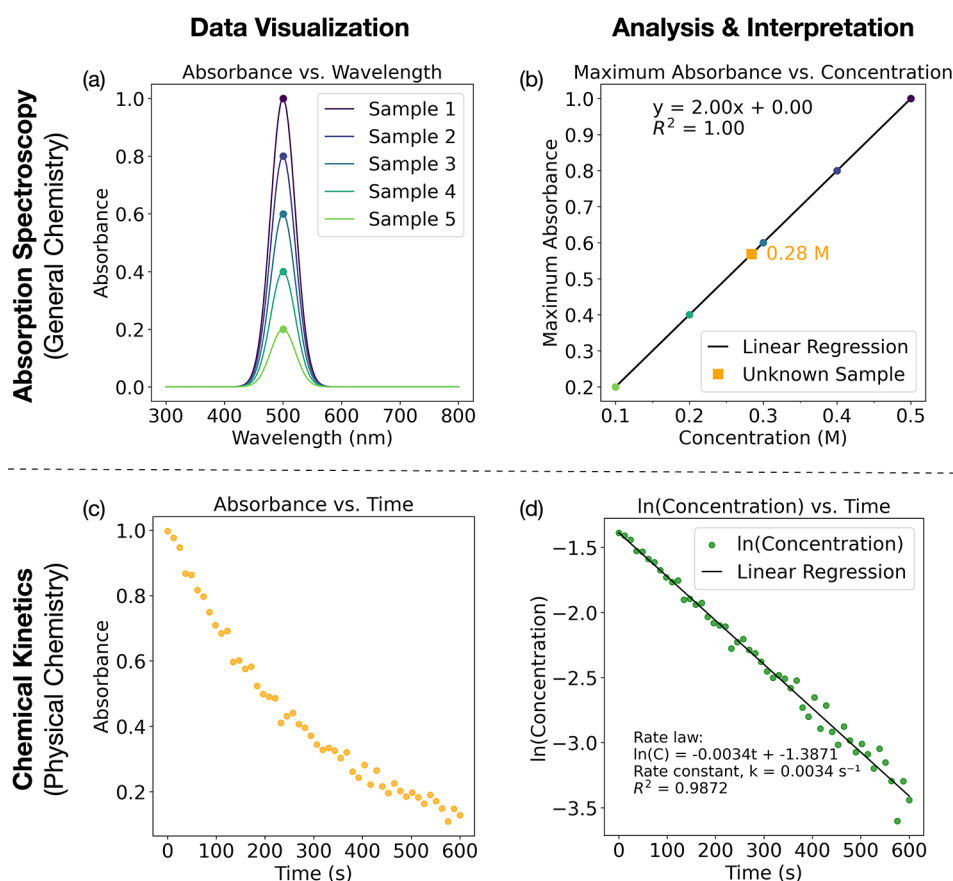


Figure 1. Demonstration of data visualization and analysis generated entirely by natural language prompting using ChatGPT-4o. The first example highlights data analysis from a general chemistry absorption spectroscopy laboratory, in which students are asked to (a) generate a calibration curve and (b) determine the concentration of an unknown analyte using Beer's law. The second example illustrates a chemical kinetics experiment familiar to upper division physical chemistry laboratories. Students (c) collect and visualize time-dependent absorbance data, relate the measured absorbance to analyte concentration with Beer's law, and then (d) linearize the data to determine the reaction order, rate constant, and integrated rate law.

laboratory arc, spanning from pre-lab assignments to written laboratory reports.

The Potential of Virtual Teaching Assistants. During the hackathon portion of the workshop, participants gravitated toward the idea of Virtual Teaching Assistants (VTAs)—LLM-based tools capable of assisting students in pre-lab preparation, in-lab troubleshooting, and post-lab assignments. These VTAs were envisioned as scalable supports that could address questions, provide feedback, and free up human TAs or instructors to focus on higher-level instructional tasks.

Pre-laboratory Preparation. Preparation for the hands-on laboratory experience is critical for student learning,^{9,10} introducing relevant concepts, calculations, and safety protocols in advance. In our hackathon, teams explored several approaches to incorporating LLM-based VTAs into these activities, aiming to help students arrive at the lab more confident and better informed.

Since many of the workshop participants were relatively new to LLMs (we imagine that some of our readers might be, as well), we included a quick primer on key terms that frequently arise in discussions about LLMs. The first concept, *Agents*, describes AI-driven systems that can autonomously recall past events, plan future action, and carry out tasks. Such tasks could involve physical lab hardware or, more commonly, other software such as executing a Python interpreter. *Retrieval-Augmented Generation (RAG)* refers to integrating external

knowledge bases into an LLM's response process to ensure more context-specific and current outputs. *Fine-tuning* is the process of refining a pre-trained model using specialized data sets, so that its responses align more closely with specific domains or tasks. *Prompt engineering* involves crafting questions or instructions that shape an LLM's style, depth, and clarity when producing answers. Finally, *chain-of-thought reasoning* encourages an LLM to articulate the step-by-step logic for arriving at a particular conclusion. Understanding these terms and their limitations proved critical in both the brainstorming and hackathon parts of the workshop.

If prompted by a student, an off-the-shelf LLM would simply hand over answers to the pre-lab assignment. During the workshop, one group developed a tool to avoid this scenario: an "Agent-Supported Pre-lab Assistant" that would provide guidance and feedback but would not provide the solution. This was accomplished by training the LLM on example student responses and acceptable feedback therein; hidden prompting further encouraged the AI assistant to guide the student toward the correct answers. Finally, evolving student responses were recorded to a transcript that was accessible to both the LLM and the instructor; understanding the progression of student learning could help instructors revise course content to effectively address misunderstandings. Another group created a "Study Buddy" tool using RAG, which allowed the LLM to pull information from a user-

defined database—such as instructor-uploaded course content—in real time. The goal was to reduce hallucinations by tailoring the LLM's domain knowledge to the exact curriculum. In the prototype, if a student answered a pre-lab quiz incorrectly, the LLM redirected them to relevant sections of the course materials and offered feedback based upon the provided course content. Together, these early prototypes demonstrate how LLMs—when combined with Agents and RAG—could enhance the pre-lab experience by offering context-specific guidance, iterative feedback, and resources tied to instructor-defined content.

Student Learning In-Lab. Participants also envisioned VTAs supplementing human TAs during laboratory sessions. They envisioned VTAs that could help students troubleshoot equipment malfunctions, adapt experimental calculations on-the-fly, or confirm safe laboratory practices. In one workshop project, a team created a custom GPT tool that used RAG to assist with these practical demands. By training the model on equipment standard operating procedures, user manuals, and sample laboratory worksheets, they ensured that the LLM would retrieve accurate, domain-specific information when students needed step-by-step guidance or targeted safety advice. In this way, the tool was confined to a creator-defined scope of knowledge to offer accurate support on-the-fly. Ultimately, such an approach illustrates the potential for LLM-driven VTAs to enhance real-time problem-solving and free human TAs to engage more deeply with students' conceptual understanding.

Post-laboratory Assignments. Post-lab activities often ask students to reflect on their results, connect experimental data to core chemical concepts, and develop essential communication skills. Data workup and interpretation, central to chemistry laboratories, require not only conceptual understanding but also practical knowledge of various software tools (e.g., Excel, Python, MATLAB). In the workshop, one group explored pre-built LLM-based tools like ChatGPT's "Data Analyst" plug-in to help visualize and analyze experimental data. By simply describing the desired plot or calculation in words, students could produce graphs without first mastering a separate coding language. This approach lowers technical barriers and enables students to focus on interpreting rather than just generating data representations.¹¹ Moreover, advanced LLMs can potentially handle complex multidimensional data, helping students tackle more sophisticated analyses than standard x-y plots. Students can now both access the underlying code used to generate these plots and use natural language coding to alter this code or integrate data analysis, all within the LLM platform.

To illustrate, we used ChatGPT-4o to generate Python scripts to analyze data for several common chemistry laboratory experiments, including (1) General Chemistry – Absorbance Spectroscopy and (2) Physical Chemistry – Reaction Kinetics. To simulate the experience of a student with no coding knowledge, we relied entirely on iterative prompting in natural language; we did not edit the final outputs from ChatGPT or provide any code initially or during prompting.

We first asked ChatGPT to generate a set of absorbance spectra following a Gaussian profile for five solutions with linearly varying analyte concentrations. We then prompted ChatGPT to visualize the absorbance spectra and determine maximum absorbance values at a given wavelength (Figure 1a). Through further prompting, ChatGPT constructed a calibra-

tion curve from the maxima in the absorbance data and performed a linear regression to determine the Beer's law relationship for the analyte. Using this calibration curve, we then provided an "experimental" absorbance value for an analyte solution of unknown concentration and instructed ChatGPT to determine the concentration (shown as an orange square in Figure 1b). This example highlights how students with minimal coding experience could potentially expand their data analysis and interpretation capabilities using LLMs.

In upper division chemistry laboratories, the data analysis is often more complex, and thus the opportunities are even more compelling. Chemical kinetics is an experiment that is routinely performed in physical chemistry. In this experiment, students are tasked with (i) developing an absorbance calibration curve for the analyte of interest, (ii) measuring the absorbance of an analyte over time, (iii) relating absorbance to changing concentration of analyte, and (iv) determining an integrated rate law and rate constant for the reaction. In Figure 1c, we used ChatGPT-4o to produce and visualize a data set of absorbance vs time during the course of a chemical reaction. Using a toy calibration curve similar to Figure 1a, we prompted ChatGPT-4o to convert the absorbance values to concentrations, plot the data as $\ln(\text{concentration})$ vs time, incorporate a linear regression, and determine the rate law for the reaction (Figure 1d). ChatGPT-4o quickly facilitates this complex, multistep analysis, allowing the user to refine and update the analysis in an iterative fashion with no prior coding knowledge.

During these tests, we also integrated conceptual questions into the prompt—e.g., "Explain Beer's law?" "What are the units of the rate constant?" "Is this a first- or second-order reaction?" For these relatively straightforward examples, ChatGPT-4o provided reasonably accurate conceptual explanations of the topic, as well as explanations of relevant mathematical relationships. Integrating data interpretation with real-time conceptual feedback in a single LLM tool could provide support for students to improve their understanding of the data in the context of chemistry concepts. Through these two examples, we found that prompting ChatGPT in natural language allowed the user to develop, in their own words, their understanding of the data visualization and analysis process without the technical barriers of programming syntax. For students who are interested in learning the underlying code, detailed code comments and conceptual explanations of the syntax can be requested during prompting. The integration of a dialogue window and a Python integrated development environment (IDE) into a single platform, as is found in ChatGPT-4o, is a powerful platform for onboarding students into programming for data analysis and visualization.

Written reports are the final stage of most laboratory exercises and challenge students to critically assess their data in narrative format and compare to theoretical predictions or literature values. Many undergraduates struggle both with the structure of formal lab reports and with technical writing in general. Because writing is inherently iterative, an LLM could assist in outlining a coherent argument, suggesting improvements during revision, or helping non-native English speakers navigate grammar and style—allowing them to devote more attention to the scientific substance. An LLM could also guide students toward more nuanced reflections on experimental error and encourage them to look beyond the common blanket statement of "human error". Further, undergraduate students

often find the scientific literature daunting. LLMs may alleviate this by identifying relevant articles or synthesizing key takeaways; users must still verify AI-generated information for technical accuracy and be wary of fictitious literature sources. Encouraging students to cross-check references can reinforce scientific rigor and help them develop the critical eye necessary for effective research.

Risks to Student Learning. Throughout the workshop, participants did not shy away from pointing out the many potential pitfalls of LLM usage in the laboratory setting. Foremost among these are hallucinations, in which false or misleading information is presented with the typical cheery confidence of LLMs.

Hallucinations. Hallucinations in chemistry education might include conceptual inaccuracies, omission of key safety details, or incorrect guidance on experimental procedures. The potential for student over-reliance on AI-generated answers further amplifies these risks. During the hackathon portion of our workshop, participants documented cases of fictitious scientific journal articles, nonexistent Python libraries, and erroneous arithmetic—some of which were initially convincing. Such incidents raised pressing questions: To what extent can RAG and custom-trained GPTs improve the accuracy of LLM responses? What strategies of prompt engineering might reduce false information? How can we teach students to critically evaluate AI responses and detect hallucinations?

In their current state, many off-the-shelf LLMs hallucinate too frequently and too convincingly to be used without instructor vetting. For example, ChatGPT-3.5 generated convincing explanations for a set of 85 organic reaction mechanisms but with only ~28% accuracy. Prompt engineering improved the sophistication of the explanations but did not increase the accuracy of the responses.¹² The risk of hallucinations is independent of the question format (e.g., multiple choice vs free response)¹³ and occurs for both conceptual questions and quantitative calculations.¹⁴ Even with careful oversight, the probabilistic nature of these models means there is no guarantee that all errors or fabrications will be caught, as each response is generated statistically rather than being anchored to a definitive knowledge base.¹⁵

Retrieval augmented generation (RAG) models have been widely adopted to mitigate hallucinations by grounding an LLM's outputs in a user-defined database to provide domain-specific context for each response. Combining RAG with effective prompting is a powerful way to improve the accuracy of an LLM's output. Using RAG alongside machine-learning-optimized prompts significantly reduced error rates in predicting the topological surface area of molecules.¹⁶ In educational settings, instructor-defined prompt engineering or agent frameworks can further guide how the LLM responds to student queries. For example, prompting an LLM to behave as a Socratic-style tutor for general chemistry can guide students toward accurate explanations of scientific phenomena while encouraging iterative self-reflection of their own understanding.¹⁷ Yet, even with RAG, hallucinations can persist and will remain an ongoing challenge.^{18,19} Recent pedagogical efforts have capitalized upon the potential for hallucinations in an effort to improve critical thinking among chemistry students; asking students to evaluate the accuracy of LLM responses or to compare AI- and human-generated texts encouraged students to challenge their understanding and engage more deeply with the subject matter.²⁰ However, this strategy may pose difficulties for advanced or nuanced topics,

where an LLM may provide accurate (but incomplete) information and students lack the expertise to identify subtle inaccuracies. As LLMs become more commonplace in the classroom and teaching laboratory, educators will need to develop structured approaches that help students question, verify, and ultimately learn how to responsibly leverage AI-generated content.

Displacing Peer Teaching and Mentorship. Workshop participants emphasized that although AI can certainly expedite routine tasks, it also risks overshadowing the human–human exchanges that define a rich laboratory experience: personalized mentorship, collaborative problem-solving, and real-time adaptation to student needs. These deeper interactions are where learners often develop scientific identity, sharpen critical thinking, and cultivate the curiosity that sparks genuine discovery.²¹ Additionally, peer interactions in general chemistry laboratories have been previously shown to improve students' commitment to learning and attitudes toward chemistry.^{21,22} Anecdotally, we have noticed that our office hour attendance is dropping as a greater fraction of students are regularly using LLMs to answer their questions.

These concerns become even more acute when we consider the central role of peer teaching and mentorship in chemistry laboratories. Teaching assistants (TAs)—often advanced undergraduates or graduate students—are an integral part of laboratories. Furthermore, students commonly work in groups, learning from each other and building confidence through mutual support. Reducing or eliminating these peer-to-peer interactions in favor of AI assistants may inadvertently deprive learners of invaluable collaborative experiences. With that said, student TAs are not infallible; they may be stretched thin by their own academic workload or inadvertently pass along misunderstandings. Taken together, these observations suggest that mitigating the risks posed by both human- and AI-generated misinformation will likely require a carefully balanced, hybrid approach that leverages the strengths of emerging technologies without displacing the interpersonal exchanges that enrich laboratory learning.

Other Topics. Participants also voiced concerns about over-reliance on AI-driven answers, especially when it might displace hands-on problem-solving, mentorship, and scientific inquiry. Issues related to academic integrity and the difficulty of detecting AI-generated work were also raised. Equity questions surfaced, particularly for institutions that may lack the necessary computing resources or students with limited access to LLMs.

2. COURSE DELIVERY STRATEGIES

While much of our discussion has focused on how LLMs influence students (through, e.g., VTAs, pre-lab preparation, or post-lab analyses), workshop participants also identified a range of ways these tools could support instructors in designing and delivering courses. In the chemistry laboratory, instructors often juggle multiple responsibilities: crafting engaging experiments, training TAs, ensuring safety compliance, and tailoring course content for students with diverse backgrounds and learning needs. Custom-trained LLMs present an opportunity to streamline these logistical tasks. Recently, Clark et al. demonstrated that ChatGPT-4 excels at supporting general chemistry instructors in course preparation tasks such as outlining a lesson plan, offering explanations and teaching strategies, and generating assessments.²³ ChatGPT-4 can also generate and grade chemistry exams with only modest

inaccuracies.²⁴ At our workshop, one group developed a prototype system in which instructors upload a set of standard operating procedures and safety documentation; the LLM could then generate consistent, lab-specific training modules for TAs. By testing these materials with new TAs—akin to a scenario-based AI “simulation”—the workshop group proposed that human TAs could be better prepared for real-time troubleshooting and safety enforcement, creating a more uniform experience across multiple lab sections.

Refining Accessibility and Inclusivity in Laboratories.

In another small-group session, participants explored how LLMs might help instructors address the varied needs of students with physical disabilities, cognitive or learning differences, or language barriers. Most instructors are not experts in accessible lab design; elements like adjustable bench setups, appropriate labeling for color-blind students, or clearly scaffolded instructions for neurodiverse learners can be overlooked. An LLM “consultant” that digests institutional guidelines or open-source accessibility resources could suggest simple but impactful adaptations. An advanced prototype developed in the hackathon proposed letting an LLM parse a given lab protocol and highlight opportunities for inclusive design, such as specifying extra safety cues for hearing-impaired students or offering alternative instructions in plain language. In principle, this assistance could help instructors better anticipate challenges, provide uniform support materials to TAs, and foster an environment where students’ diverse needs are addressed more thoroughly.

Setting-Agnostic Course Support and Ethics Integration. Beyond lab-specific considerations, participants also noted how LLMs could streamline broader course delivery tasks, from creating custom curricular materials to assessing student performance. Some groups toyed with the idea of an AI-driven “diagnostic” that analyzes class-wide quiz data, flags common misconceptions, and even suggests supplementary problem sets. Especially in larger classes, this could help instructors tailor their pacing or revisit tough topics more effectively. Another recurring theme was the integration of ethical discourse into the chemistry curriculum: given that modern research increasingly relies on AI-driven discovery methods (e.g., drug design, materials informatics), the lab offers an ideal environment to have conversations about algorithmic bias, data privacy, and responsible AI use. As one team emphasized, if students are already interacting with LLMs for data interpretation or research prompts, instructors can use that engagement as a jumping-off point to discuss the ethical dimensions of letting AI shape future scientific breakthroughs. Ultimately, instructor time and energy are finite resources, and LLMs are well-structured to help shoulder some of the load.

3. ALTERNATIVE LABORATORY STRUCTURES

In contrast to traditional laboratories, inquiry-based laboratories and course-based undergraduate research experiences (CUREs) can elevate students’ higher-order thinking, build confidence, and nurture problem-solving abilities.^{25–29} LLMs may help make such alternative approaches more feasible, emphasizing student autonomy and critical thinking over rote procedure-following.

One way to embrace inquiry is through computationally intensive laboratories. Although there are rapidly growing efforts to incorporate computationally driven laboratories into the chemistry and materials science curricula, the practical implementation has been quite variable across institutions and

programs.^{30,31} For example, a course might introduce density functional theory (DFT) simulations or code-based modeling of chemical or physical phenomena. Typically, many undergraduate students lack proficiency in scripting languages like Python or MATLAB, which can impede large-scale adoption of computational projects. Here, AI tools excel at generating and debugging code in response to plain-language prompts (as illustrated in Figure 1, above), allowing students to spend less time wrestling with syntax and more time interpreting scientific results. By minimizing barriers to coding, LLMs may help instructors scale computational laboratories to larger groups and encourage deeper exploration of core concepts.

CUREs provide another avenue for student-centered research, though implementing open-ended projects in courses of 30 or 200 students is notoriously demanding. Problems like instrument availability, safety protocols, and grading complex projects can deter many departments from adopting such models. Yet LLMs might ease some of these burdens by suggesting experiments, guiding students through iterative data analysis, or providing quick feedback on project proposals. AI-driven assistants could also highlight relevant literature or help students synthesize conclusions from multiple sources—helpful for students who are new to research—while leaving instructors free to mentor the deeper aspects of inquiry. Of course, these tools will not resolve every obstacle, and caution is needed to prevent over-reliance or uncritical acceptance of AI outputs. However, pairing the motivational benefits of authentic research experiences with the scaffolding afforded by generative AI may open doors for more accessible, large-scale adoption of CUREs or similar inquiry-based formats in undergraduate curricula.

CONCLUSIONS AND OUTLOOK

AI and LLMs are here to stay, and they are evolving at a pace that challenges even the most forward-thinking educators. As these tools grow more powerful and nuanced, our strategies for teaching chemistry (and beyond) must likewise adapt. The NSF-funded workshop described here sought to demonstrate a range of novel ways in which AI could enhance learning in the laboratory, and participants showcased the creative potential of approaches that pair generative models with tools like Agents and Retrieval-Augmented Generation. These innovations highlight just how dynamic and flexible LLM-based tools can be when thoughtfully integrated into chemistry curricula. While we caution that these tools are no panacea—they require vigilant oversight, robust prompt engineering, and critical thinking skills among learners—workshop participants generally agreed that AI can be a net benefit when approached conscientiously.

Finally, our workshop explored how current iterations of LLMs could be incorporated into *existing* course formats to impact both teaching and learning experiences. However, the rapid evolution of LLMs may quickly disrupt traditional instructional paradigms, fundamentally transforming how we conceptualize classroom dynamics, laboratory instruction, and even the roles of students and educators. Students will need to navigate an evolving educational landscape that integrates these systems and increasingly take ownership of their own learning. Preparing for this potential future will require educators not only to adapt to technological advancements but also to rethink pedagogical strategies and ethical considerations. A crucial lesson from our workshop is the need for a community of educators to emerge and share both

best practices and cautionary tales. Because LLM technology continues to evolve, our collective understanding of what works (and what does not) must also remain flexible. By collaborating across institutions and disciplines, we can pool resources, compare results, and refine pedagogical strategies in near-real-time. In many ways, predicting the long-term future of AI in education is inherently optimistic: we cannot know what new capabilities will surface next year, let alone in the next decade, but we can equip ourselves and our students with the mindset to navigate these shifts constructively.

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Notes

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